

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40012
Subject Title	Machine Learning in Physics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	AP20005 or COMP1012
Objectives	To introduce the fundamentals of machine learning and the applications in physical science.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) extract, visualize and process data from online databases dedicated to materials scientists and physicists; (b) select machine learning models to solve specific problems; (c) implement various machine learning models in Python; and (d) evaluate and improve the performance of a model.
Subject Synopsis/ Indicative Syllabus	Linear regression and classification: databases, feature, representation, training set, error functions, gradient descent, univariate and multi-variate regression, logistic regression Neural networks: activation functions, back propagation, feed-forward networks, recurrent networks, deep learning, CGCNN Theory of machine learning: overfitting, validation, regularization, generalization, bias-variance Application in physical science: Materials Project database, bulk modulus, bandgap, diffusion barrier, energy, force, molecular dynamics, Gibbs free energy
Teaching/Learning Methodology	The course will make use of public cloud computing to enhance the learning using nanoHUB.org as well as its features for collaboration and publishing. Students will learn how to program in Python with Jupyter notebooks and use external libraries such as Numpy, Scipy, Pandas and Tensorflow. Lecture: The concepts will be presented in class including some illustrative examples and algorithms related to solving problems in physics or engineering. Computer laboratory: During laboratory sessions, students will write small programs individually or in groups to gain a deeper understanding of the topics discussed during the lectures. These sessions aim at developing the students' intuition and critical thinking upon discussion and exchange of ideas in order to engage in the proposed activities.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	(1) Continuous assessment	20	√	√	√	√
	(2) Project	15	√	√	√	√
	(3) Test	15		√		√
	(3) Final exam	50		√		√
	Total	100 %				
Continuous assessment includes participation in the computer laboratory sessions and turning on time the graded quizzes in order to check students' progress throughout the semester. The project consists of project presentations and project report and will be assigned during the second half of the semester. Students will use their knowledge gained in the class/lab to design a machine learning program in order to tackle a problem related to physical science. Presentations and reports will be graded. Test contains various questions which are designed to access the learning outcomes for the first half semester. The final examination is a 2-hour written final examination. Various questions will be set in both components to assess the intended learning outcomes.						
Student Study Effort Expected	Class contact:					
	• Lecture			26 Hrs.		
	• Computer Laboratory			13 Hrs.		
	Other student study effort:					
	• Self-study			81 Hrs.		
	• Total student study effort			120 Hrs.		
Reading List and References	<i>Hands-on machine learning with Scikit-Learn and TensorFlow: concepts, tools, and techniques to build intelligent systems.</i> Géron, Aurélien. O'Reilly Media, Inc., 2017.					
	<i>A first course in machine learning.</i> Rogers, Simon, and Mark Girolami. Chapman and Hall/CRC, 2016.					
	<i>Machine Learning</i> , MOOC from Stanford (CS229). Ng, Andrew.					
	<i>Learning From Data</i> , MOOC from Caltech. Yaser S. Abu-Mostafa.					

